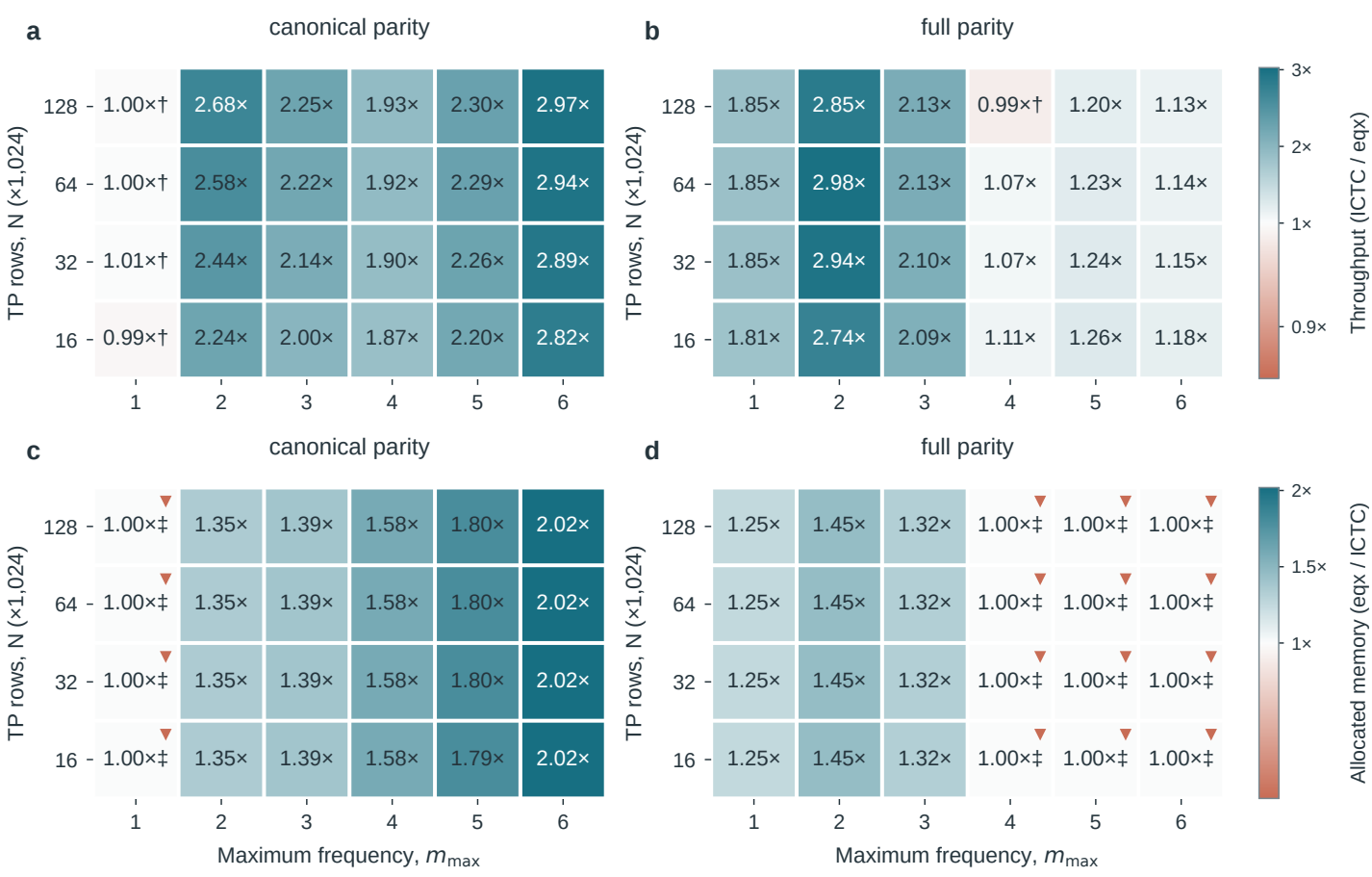

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward · C = 128 · N = 16,384–131,072 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
TP rows, N (×1,024)
128
64
32
16
1.00×† 2.68× 2.25× 1.93× 2.30× 2.97×
1.00×† 2.58× 2.22× 1.92× 2.29× 2.94×
1.01×† 2.44× 2.14× 1.90× 2.26× 2.89×
0.99×† 2.24× 2.00× 1.87× 2.20× 2.82×
1.85× 2.85× 2.13× 0.99×† 1.20× 1.13×
1.85× 2.98× 2.13× 1.07× 1.23× 1.14×
1.85× 2.94× 2.10× 1.07× 1.24× 1.15×
1.81× 2.74× 2.09× 1.11× 1.26× 1.18×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.79× 2.02×
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1 2 3 4 5 6
Maximum frequency, m_max
Throughput (ICTC / eqx)
3×
2×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5